LAB 3: EXPLORING UDP WITH DNS USING WIRESHARK

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SECTION: C

PART 1: DNS ANALYSIS

1. Observe DNS Caching:

```
dig google.com

; <<>> DiG 9.20.17 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41434
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             199     IN      A       142.250.205.206

;; Query time: 7 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:51:35 IST 2026
;; MSG SIZE  rcvd: 55
```

```
» dig google.com

; <<>> DiG 9.20.17 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50874
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             192     IN      A       142.250.205.206

;; Query time: 20 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:51:43 IST 2026
;; MSG SIZE  rcvd: 55
```

udp.port==53

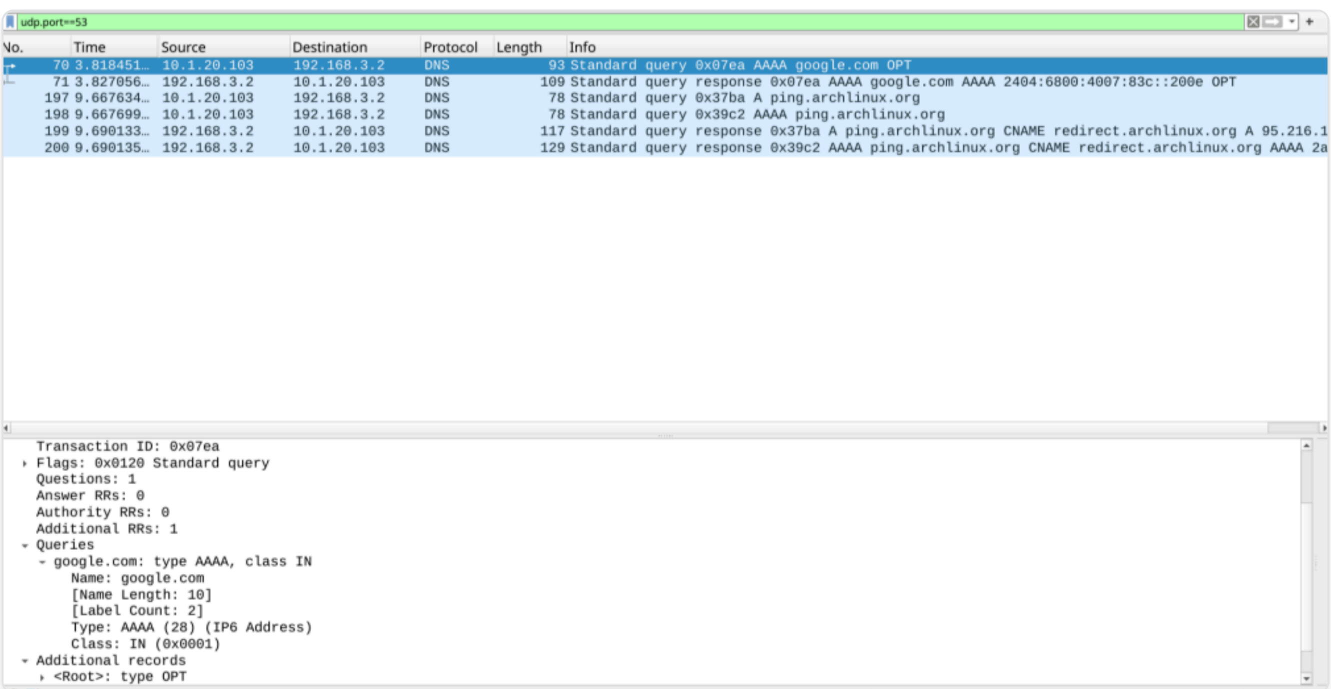
No.	Time	Source	Destination	Protocol	Length	Info
94	7.545681…	10.1.20.103	192.168.3.2	DNS	93	Standard query 0xa1da A google.com OPT
95	7.552904…	192.168.3.2	10.1.20.103	DNS	97	Standard query response 0xa1da A google.com A 142.250.205.206 OPT
224	15.04388…	10.1.20.103	192.168.3.2	DNS	93	Standard query 0xc6ba A google.com OPT
225	15.06382…	192.168.3.2	10.1.20.103	DNS	97	Standard query response 0xc6ba A google.com A 142.250.205.206 OPT
326	20.82647…	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x9893 A ping.archlinux.org
327	20.82654…	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x618c AAAA ping.archlinux.org
328	20.83234…	192.168.3.2	10.1.20.103	DNS	117	Standard query response 0x9893 A ping.archlinux.org CNAME redirect.archlinux.org A 95.216.1
329	20.83234…	192.168.3.2	10.1.20.103	DNS	129	Standard query response 0x618c AAAA ping.archlinux.org CNAME redirect.archlinux.org AAAA 2a

```
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
  ▾ Queries
    ▾ google.com: type A, class IN
        Name: google.com
        [Name Length: 10]
        [Label Count: 2]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
  ▾ Additional records
    ▸ <Root>: type OPT
```

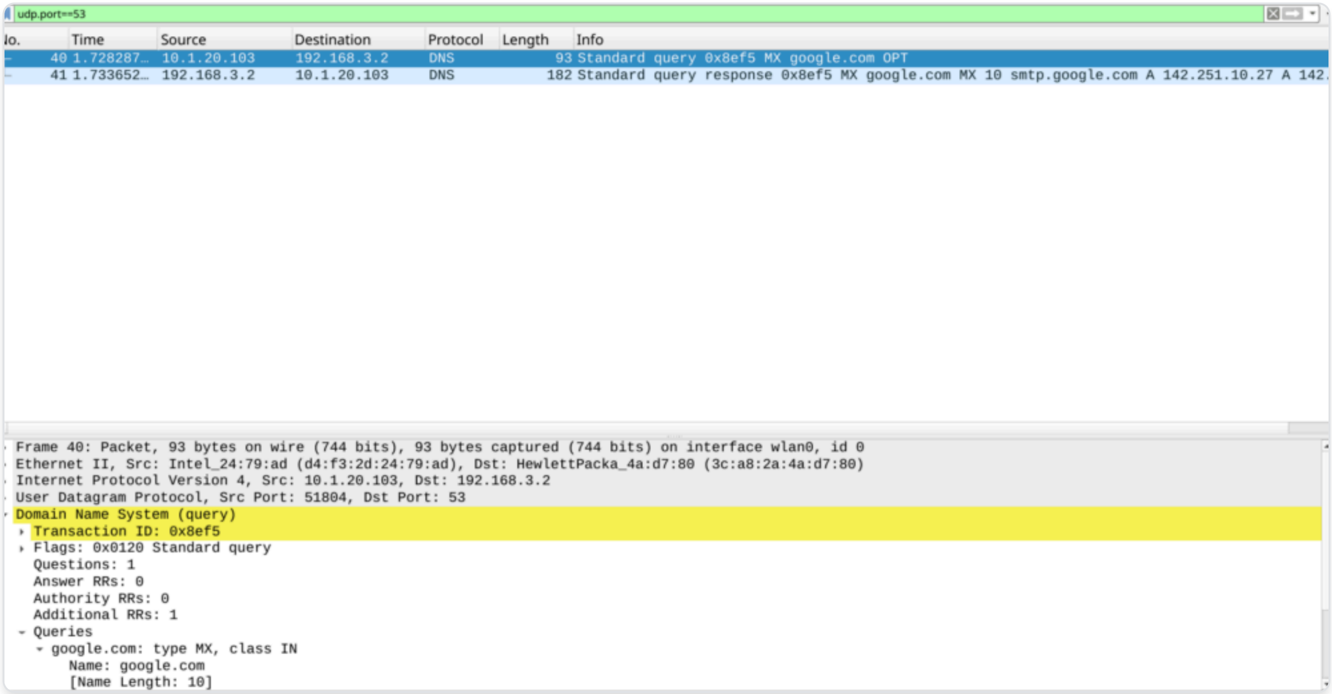
2.Query for Other Record Types:

try `dig www.google.com AAAA` to observe the IPv6 address.

```
dig google.com AAAA

; <<>> DiG 9.20.17 <<>> google.com AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2026
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                    IN      AAAA

;; ANSWER SECTION:
google.com.             205     IN      AAAA    2404:6800:4007:83c::200e

;; Query time: 9 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:29:11 IST 2026
;; MSG SIZE  rcvd: 67
```



Use `dig google.com MX` to view mail server records.



```
dig google.com MX

; <<>> DiG 9.20.17 <<>> google.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44515
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 6

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                    IN      MX

;; ANSWER SECTION:
google.com.             205     IN      MX      10 smtp.google.com.

;; ADDITIONAL SECTION:
smtp.google.com.        148     IN      A       142.251.12.26
smtp.google.com.        148     IN      A       142.251.12.27
smtp.google.com.        148     IN      A       142.251.10.26
smtp.google.com.        148     IN      A       64.233.170.27
smtp.google.com.        148     IN      A       142.251.10.27

;; Query time: 6 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:46:33 IST 2026
;; MSG SIZE  rcvd: 140
```

3. Modify DNS Servers:

```
❯ dig @1.1.1.1 www.google.com

; <<>> DiG 9.20.17 <<>> @1.1.1.1 www.google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49018
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.google.com.                        IN      A

;; ANSWER SECTION:
www.google.com.         42      IN      A       142.251.221.228

;; Query time: 58 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Thu Feb 12 14:58:17 IST 2026
;; MSG SIZE  rcvd: 59
```

```
dig @8.8.8.8 www.google.com

; <<>> DiG 9.20.17 <<>> @8.8.8.8 www.google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3933
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.google.com.                        IN      A

;; ANSWER SECTION:
www.google.com.         179     IN      A       142.251.221.164

;; Query time: 17 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Thu Feb 12 14:58:29 IST 2026
;; MSG SIZE  rcvd: 59
```

› **Cloudflare: 58ms**

› **Google: 17ms**

udp.port==53

No.	Time	Source	Destination	Protocol	Length	Info
141	4.704161…	10.1.20.103	1.1.1.1	DNS	97	Standard query 0xbf7a A www.google.com OPT
150	4.762275…	1.1.1.1	10.1.20.103	DNS	101	Standard query response 0xbf7a A www.google.com A 142.251.221.228 OPT
474	16.23039…	10.1.20.103	8.8.8.8	DNS	97	Standard query 0x0f5d A www.google.com OPT
475	16.24771…	8.8.8.8	10.1.20.103	DNS	101	Standard query response 0x0f5d A www.google.com A 142.251.221.164 OPT

```
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
  ▾ Queries
    ▾ www.google.com: type A, class IN
        Name: www.google.com
        [Name Length: 14]
        [Label Count: 3]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
  ▾ Additional records
    ▸ <Root>: type OPT
    [Response In: 150]
```

4. Analyze DNS over TCP:

```
dig +tcp www.google.com

; <<>> DiG 9.20.17 <<>> +tcp www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57961
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.google.com.                        IN      A

;; ANSWER SECTION:
www.google.com.         100     IN      A       142.251.220.4

;; Query time: 4 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (TCP)
;; WHEN: Thu Feb 12 15:10:23 IST 2026
;; MSG SIZE  rcvd: 59
```

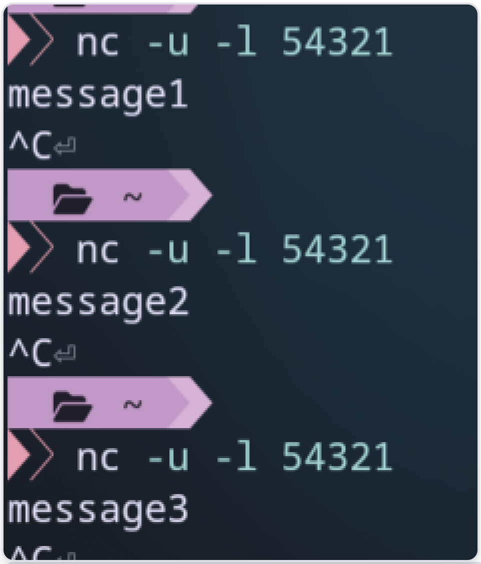
tcp.port==53

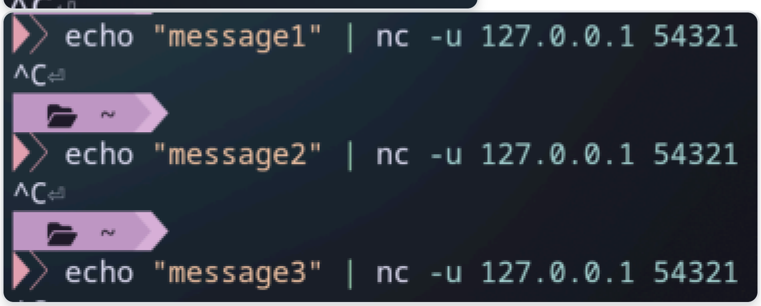
No.	Time	Source	Destination	Protocol	Length	Info
67	1.969555…	10.1.20.103	192.168.3.2	TCP	74	45723 → 53 [SYN] Seq=0 Win=64660 Len=0 MSS=1220 SACK_PERM TSval=3505431216 TSecr=0 WS=1024
68	1.974328…	192.168.3.2	10.1.20.103	TCP	74	53 → 45723 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=27636677
69	1.974386…	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3505431221 TSecr=276366777
70	1.974509…	10.1.20.103	192.168.3.2	DNS	123	Standard query 0xe269 A www.google.com OPT
71	1.979779…	192.168.3.2	10.1.20.103	DNS	127	Standard query response 0xe269 A www.google.com A 142.251.220.4 OPT
72	1.979831…	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=58 Ack=62 Win=65536 Len=0 TSval=3505431226 TSecr=276366777
73	1.980674…	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [FIN, ACK] Seq=58 Ack=62 Win=65536 Len=0 TSval=3505431227 TSecr=276366777
74	1.983936…	192.168.3.2	10.1.20.103	TCP	66	53 → 45723 [ACK] Seq=62 Ack=59 Win=2098176 Len=0 TSval=276366786 TSecr=3505431227
75	1.983936…	192.168.3.2	10.1.20.103	TCP	66	53 → 45723 [FIN, ACK] Seq=62 Ack=59 Win=2098176 Len=0 TSval=276366786 TSecr=3505431227
76	1.984006…	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=59 Ack=63 Win=65536 Len=0 TSval=3505431230 TSecr=276366786

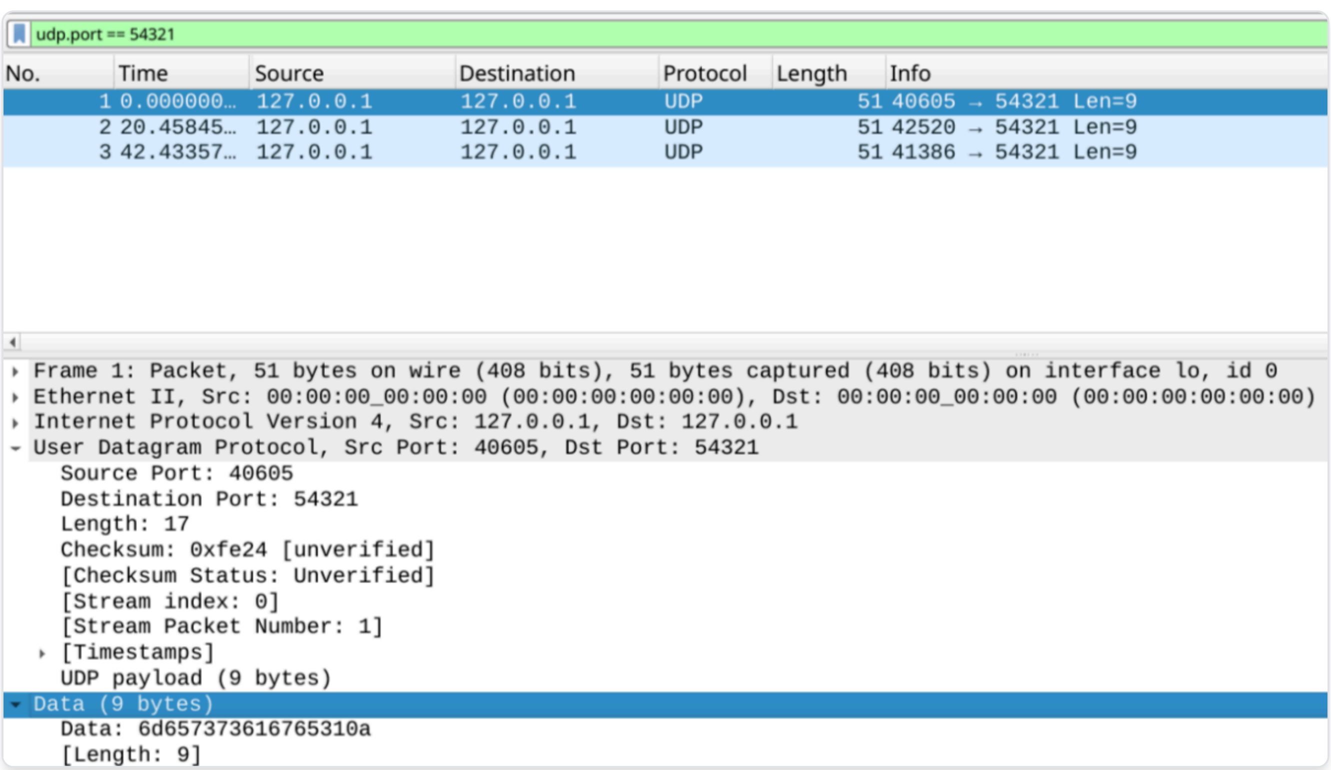
```
Frame 67: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)
Internet Protocol Version 4, Src: 10.1.20.103, Dst: 192.168.3.2
Transmission Control Protocol, Src Port: 45723, Dst Port: 53, Seq: 0, Len: 0
    Source Port: 45723
    Destination Port: 53
    [Stream index: 0]
    [Stream Packet Number: 1]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 1252856957
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1010 .... = Header Length: 40 bytes (10)
    Flags: 0x002 (SYN)
    Window: 64660
    [Calculated window size: 64660]
    Checksum: 0xe240 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
    [Timestamps]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

PART 2: UDP COMMUNICATION

> **Set up a UDP server using netcat on port 54321.**







udp.port == 54321

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000…	127.0.0.1	127.0.0.1	UDP	51	40605 → 54321 Len=9
2	20.45845…	127.0.0.1	127.0.0.1	UDP	51	42520 → 54321 Len=9
3	42.43357…	127.0.0.1	127.0.0.1	UDP	51	41386 → 54321 Len=9

```
Frame 1: Packet, 51 bytes on wire (408 bits), 51 bytes captured (408 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
User Datagram Protocol, Src Port: 40605, Dst Port: 54321
    Source Port: 40605
    Destination Port: 54321
    Length: 17
    Checksum: 0xfe24 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
    [Timestamps]
    UDP payload (9 bytes)
Data (9 bytes)
    Data: 6d657373616765310a
    [Length: 9]
```

Write a report:

› How does UDP differ from TCP?

UDP is connectionless and does not guarantee delivery, ordering, or error recovery. It has a fixed 8 byte header and no handshake. TCP is connection-oriented, reliable, ensures ordered delivery, and uses flow and congestion control, but with higher overhead.

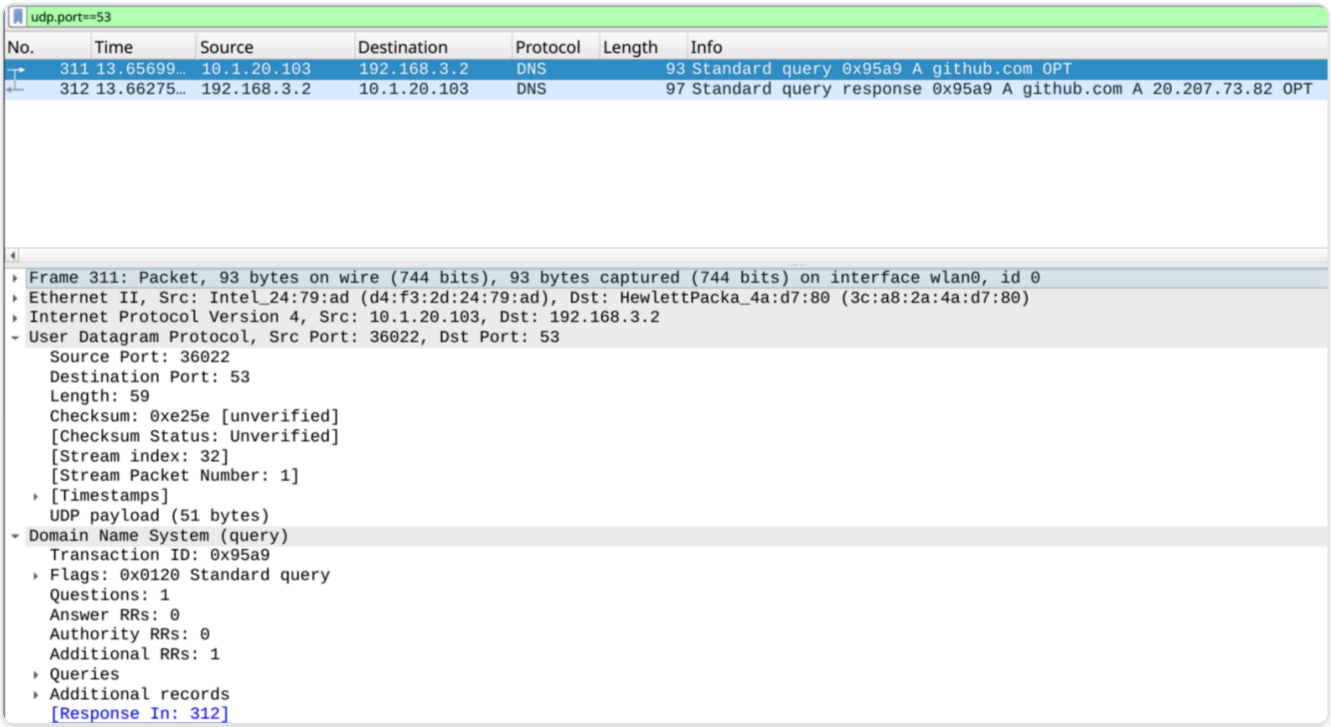
› Why is UDP suitable for DNS?

DNS queries are small and require fast responses. UDP avoids connection setup delays, reduces overhead, and allows DNS servers to handle large volumes of short request-response transactions efficiently. TCP is only used when responses are large or for zone transfers.

› Observations about UDP traffic in Wireshark:

UDP packets show only four header fields (Source Port, Destination Port, Length, Checksum). No handshake packets are seen. In DNS traffic, the client uses an ephemeral source port and sends to port 53, and the server reply reverses the port numbers.

PART 3: ANALYSIS OF THE UDP PROTOCOL IN DNS USING WIRESHARK



1. Select the first UDP segment in your trace

› 1. What is the packet number of this segment in the trace file?

- **PACKET NUMBER:** 311

› 2. What type of application-layer payload or protocol message is being carried in this UDPsegment? Look at the details of this packet in Wireshark.

- **PROTOCOL CARRIED: DNS STANDARD QUERY**

› 3. How many fields are there in the UDP header? What are the names of these fields?

› UDP header fields: 4

Source port

Destination port

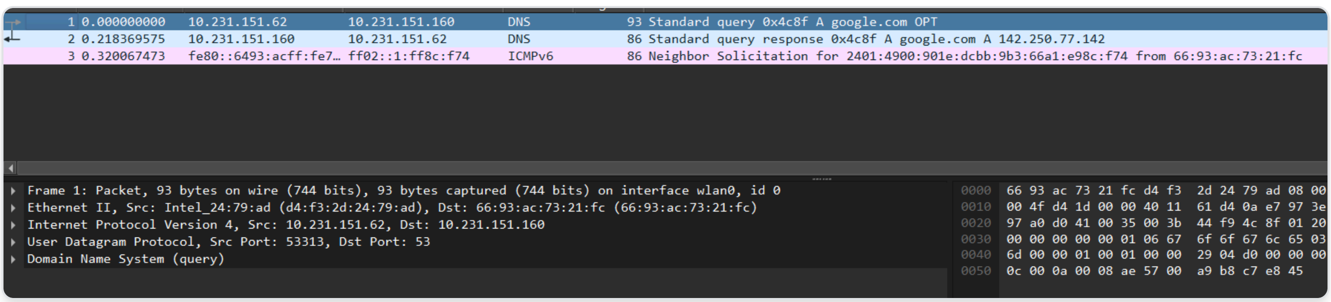
Length

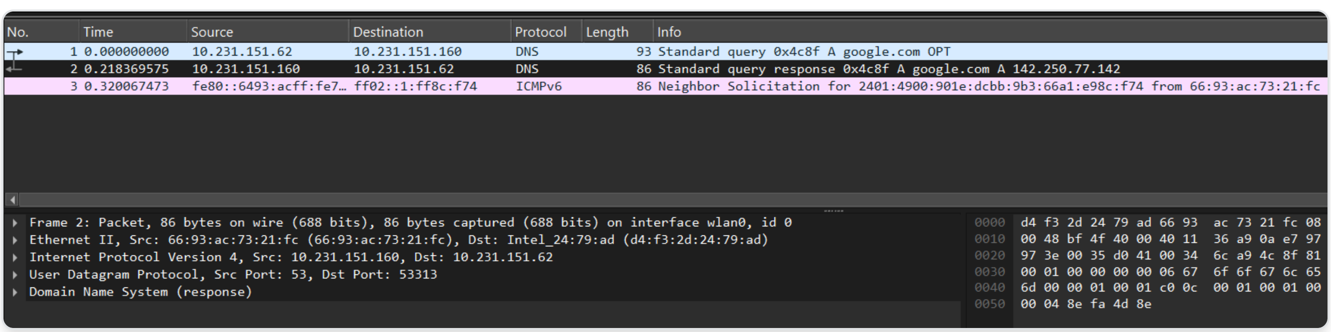
Checksum

> ## 2. By consulting the displayed information in Wireshark's packet content field for this packet, what is the length (in bytes) of the UDP header field?

- UDP length (59) - UDP payload(51) -> 8bytes.

> ## 3. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet.





› A. What is the packet number of these two UDP segments in the trace file?

› 1 and 2

› B. What is the value in the source port field in this UDP segment?

- 53313

› C. What is the value in the destination port field in this UDP segment?

- 53

› D. What is the value in the source port field in this second UDP segment?

- 53

› E. What is the value in the destination port field in this second UDP segment? Describe the relationship between the port numbers in the two packets

- 53313

- THE CLIENT PICKS A TEMPORARY PORT, AND THE SERVER REPLIES BACK TO THAT SAME PORT.